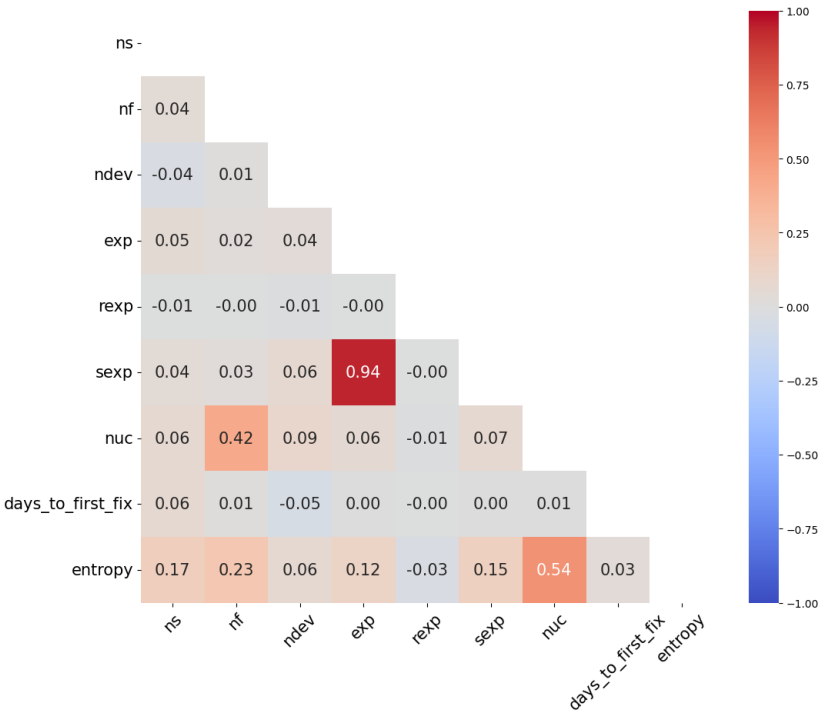


Figure A1: Heat map of correlation coefficients of Rust dataset

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Appendix

A. Dataset Description

A heat map is a data visualisation that represents different data intensities within a dataset using colour gradients. It is often used to identify correlations, patterns, and trends in data, making it easier to display complex information quickly. It conveys potential relationships between the variables shown on the x- and y-axes by using different colors, or varying shades of the same color, to indicate various values. Darker hues or shades typically indicate larger or more quantities of the values shown in the heat map. The reddish color represents the positive correlation, and the bluish color represents the negative correlation among the variables.We will also assess distribution shapes, variability, and central tendencies using descriptive statistics. It also enable us to identify outliers and unusual values. In the below sections, we have visualized each software projects using correlation metrics and descriptive statistics.

A1. Rust

The heat map of the correlation coefficient among the variables of our interest for the Rust data set is shown in Fig. A1.

Table A1 shows the statistics that can help to understand the distribution and variability of the different variables in the Rust dataset.

A2. TensorFlow

The heat map of the correlation coefficient among the variables of our interest for the Tensorflow data set is shown in Fig. A2.

Table A2 shows the statistics that can help to understand the distribution and variability of the different variables in the Tensorflow dataset.

Table A1
Descriptive statistics of the model variables for Rust: defect-inducing commits after removal of the right-censored tail ($n = 1,422$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	145.89	53.00	178.56	(136.61, 155.17)
`entropy`	Treatment	1.47	1.09	1.55	(1.39, 1.55)
`sexp`	Treatment	3,970.4	910.00	6,796.9	(3,617.1, 4,323.7)
`rexp`	Treatment	41.16	1.10	332.96	(23.86, 58.47)
`days_to_first_fix`	Outcome	177.60	54.71	296.16	(162.21, 193.00)
`exp`	Confounder	4,167.2	979.50	7,072.6	(3,799.6, 4,534.9)
`ns`	Confounder	1.18	1.00	0.468	(1.15, 1.20)
`nf`	Confounder	15.39	3.00	71.06	(11.69, 19.08)
`lt`	Confounder	639.88	390.25	790.41	(598.79, 680.96)
`nuc`	Confounder	569.55	146.00	1,696.8	(481.35, 657.74)
`la`	Confounder	327.02	47.00	2,000.8	(223.03, 431.01)
`fix`	Confounder	0.259	0.00	0.438	(0.236, 0.282)
`age`	Mediator	18.68	3.75	46.99	(16.24, 21.13)

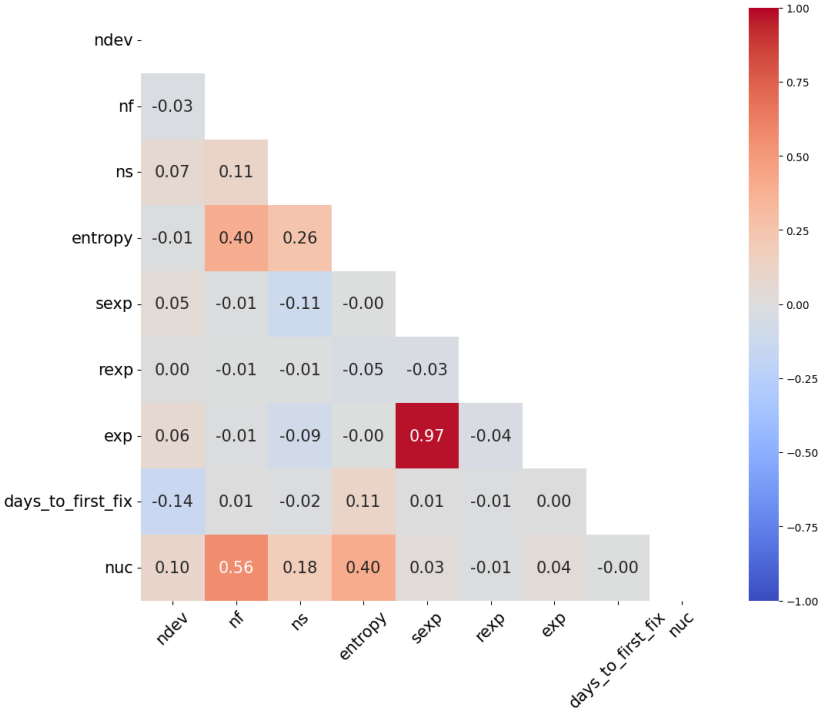


Figure A2: Heat map of correlation coefficients of Tensorflow dataset

A3. Rails

The heat map of the correlation coefficient among the variables of our interest for the Rails data set is shown in Fig. A3.

Table A3 shows the statistics that can help to understand the distribution and variability of the different variables in the Rails dataset.

Table A2
Descriptive statistics of the model variables for TensorFlow: defect-inducing commits after removal of the right-censored tail ($n = 12,926$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	201.63	199.00	111.43	(199.71, 203.56)
`entropy`	Treatment	1.51	1.31	1.25	(1.49, 1.54)
`sexp`	Treatment	19,009	817.00	30,373	(18,486, 19,533)
`rexp`	Treatment	7.94	1.09	195.88	(4.56, 11.32)
`days_to_first_fix`	Outcome	114.66	46.97	166.51	(111.79, 117.53)
`exp`	Confounder	19,384	944.50	30,627	(18,856, 19,912)
`ns`	Confounder	1.10	1.00	0.418	(1.09, 1.11)
`nf`	Confounder	9.83	4.00	51.95	(8.94, 10.73)
`lt`	Confounder	900.54	572.75	1,416.2	(876.13, 924.95)
`nuc`	Confounder	257.91	81.00	890.39	(242.56, 273.26)
`la`	Confounder	409.10	94.00	3,414.6	(350.23, 467.97)
`fix`	Confounder	0.230	0.00	0.421	(0.223, 0.237)
`age`	Mediator	19.62	8.03	35.26	(19.01, 20.23)

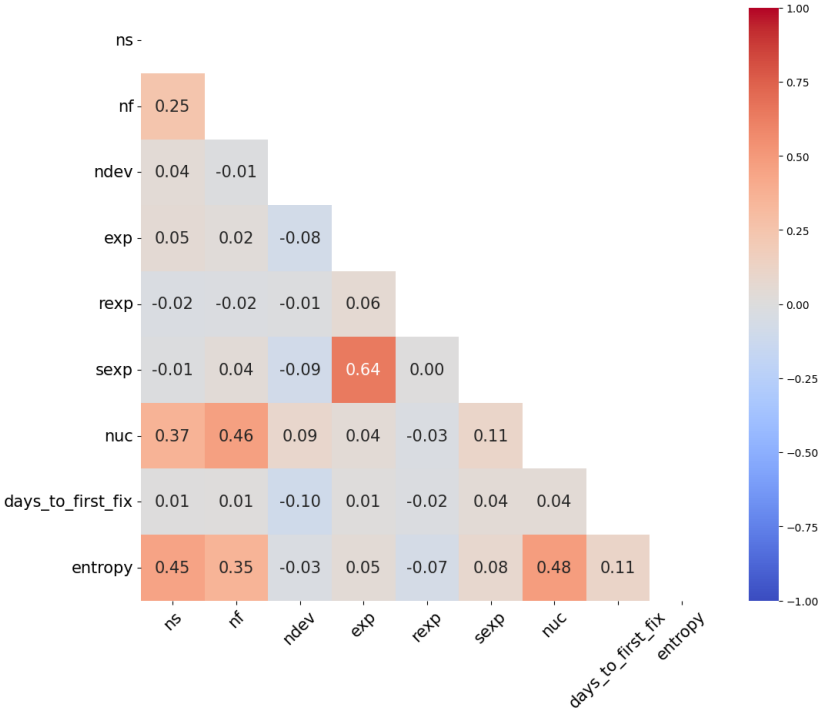


Figure A3: Heat map of correlation coefficients of Rails dataset

A4. VSCode

The heat map of the correlation coefficient among the variables of our interest for the VSCode data set is shown in Fig. A4.

Table A4 shows the statistics that can help to understand the distribution and variability of the different variables in the VSCode dataset.

Table A3
Descriptive statistics of the model variables for Rails: defect-inducing commits after removal of the right-censored tail ($n = 13,086$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	182.56	169.00	140.41	(180.15, 184.97)
`entropy`	Treatment	1.07	0.977	0.976	(1.05, 1.08)
`sexp`	Treatment	525.91	129.00	798.93	(512.22, 539.60)
`rexp`	Treatment	38.53	1.12	442.19	(30.95, 46.11)
`days_to_first_fix`	Outcome	449.77	180.25	622.91	(439.10, 460.44)
`exp`	Confounder	1,837.0	557.50	2,531.4	(1,793.6, 1,880.4)
`ns`	Confounder	1.20	1.00	0.634	(1.19, 1.21)
`nf`	Confounder	4.10	2.00	7.15	(3.98, 4.23)
`lt`	Confounder	502.93	321.43	552.25	(493.46, 512.39)
`nuc`	Confounder	317.12	127.00	461.69	(309.21, 325.03)
`la`	Confounder	81.29	21.00	927.94	(65.39, 97.19)
`fix`	Confounder	0.231	0.00	0.421	(0.224, 0.238)
`age`	Mediator	34.88	9.23	82.75	(33.46, 36.30)

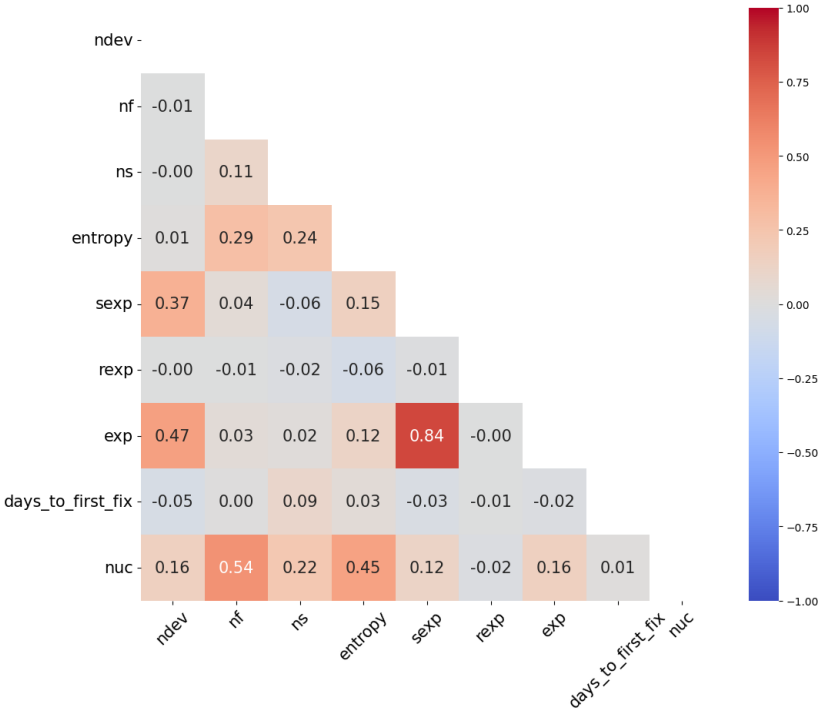


Figure A4: Heat map of correlation coefficients of VSCode dataset

A5. WP-Calypso

The heat map of the correlation coefficient among the variables of our interest for the WP-Calypso data set is shown in Fig. A5.

Table A5 shows the statistics that can help to understand the distribution and variability of the different variables in the WP-Calypso dataset.

Table A4
Descriptive statistics of the model variables for VSCode: defect-inducing commits after removal of the right-censored tail ($n = 1,023$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	42.85	43.00	22.67	(41.46, 44.24)
`entropy`	Treatment	1.23	1.03	1.15	(1.16, 1.30)
`sexp`	Treatment	2,415.1	552.00	3,872.3	(2,177.8, 2,652.4)
`rexp`	Treatment	24.50	1.23	150.33	(15.29, 33.71)
`days_to_first_fix`	Outcome	128.13	41.07	191.86	(116.37, 139.89)
`exp`	Confounder	4,757.0	3,495.0	4,927.5	(4,455.0, 5,058.9)
`ns`	Confounder	1.35	1.00	0.681	(1.31, 1.39)
`nf`	Confounder	7.45	3.00	41.91	(4.88, 10.02)
`lt`	Confounder	1,745.0	573.25	3,323.9	(1,541.4, 1,948.7)
`nuc`	Confounder	333.62	105.00	753.93	(287.42, 379.82)
`la`	Confounder	607.05	36.00	5,065.3	(296.65, 917.45)
`fix`	Confounder	0.329	0.00	0.470	(0.301, 0.358)
`age`	Mediator	16.51	3.83	39.32	(14.10, 18.92)

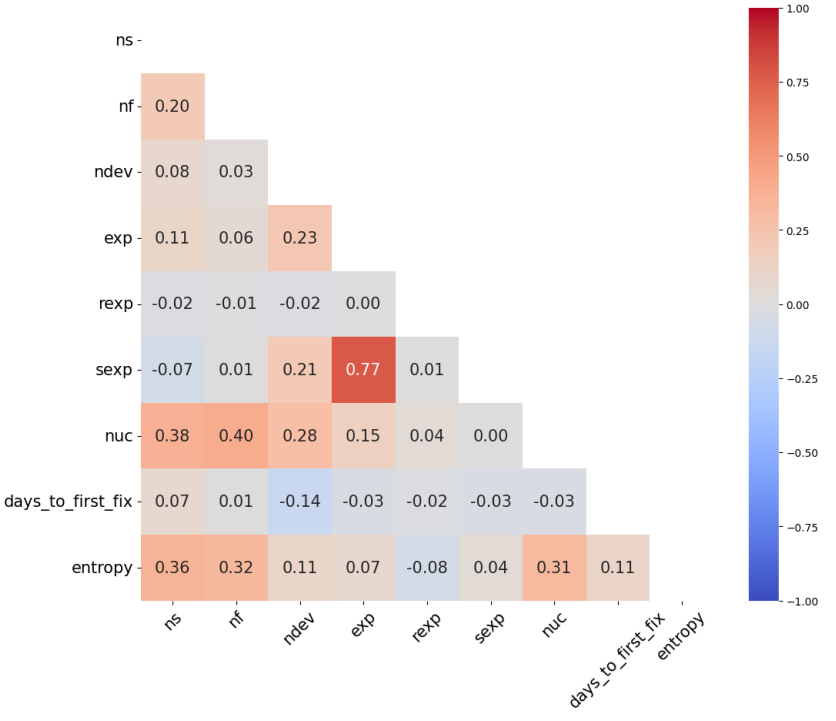


Figure A5: Heat map of correlation coefficients of WP-Calypso dataset

A6. Camel

The heat map of the correlation coefficient among the variables of our interest for the Camel data set is shown in Fig. A6.

Table A6 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table A5
Descriptive statistics of the model variables for WP-Calypso: defect-inducing commits after removal of the right-censored tail ($n = 5,596$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	108.30	113.00	57.41	(106.79, 109.80)
`entropy`	Treatment	1.32	1.15	1.15	(1.29, 1.35)
`sexp`	Treatment	421.45	201.00	651.75	(404.37, 438.53)
`rexp`	Treatment	27.59	1.07	241.86	(21.26, 33.93)
`days_to_first_fix`	Outcome	226.31	79.09	293.81	(218.62, 234.01)
`exp`	Confounder	553.35	291.25	832.25	(531.54, 575.16)
`ns`	Confounder	1.18	1.00	0.489	(1.17, 1.20)
`nf`	Confounder	6.70	3.00	45.46	(5.51, 7.89)
`lt`	Confounder	227.59	160.70	320.83	(219.18, 236.00)
`nuc`	Confounder	128.14	34.00	464.27	(115.97, 140.30)
`la`	Confounder	174.96	43.00	1,982.6	(123.01, 226.91)
`fix`	Confounder	0.255	0.00	0.436	(0.244, 0.266)
`age`	Mediator	22.05	6.51	45.17	(20.87, 23.23)

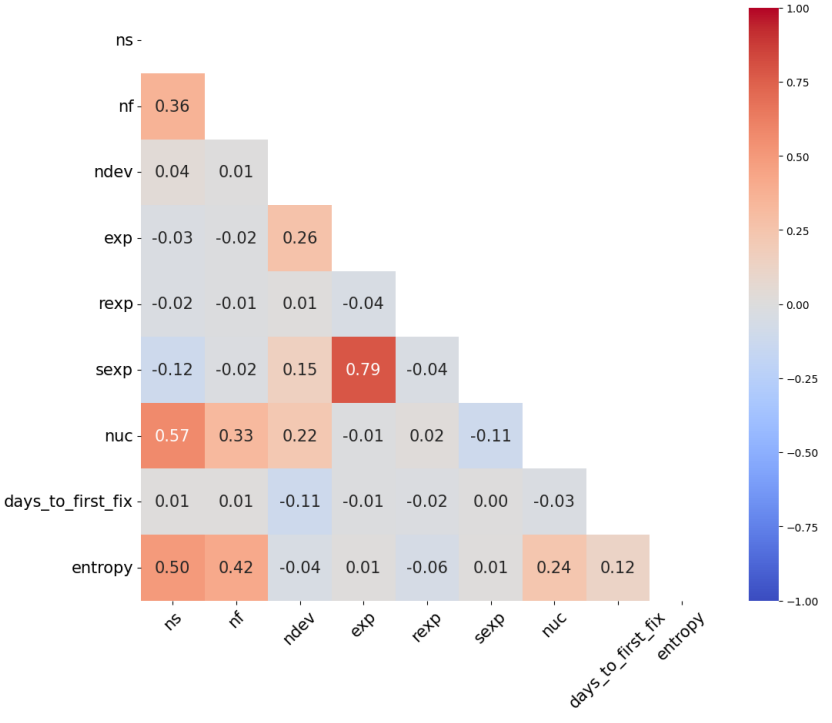


Figure A6: Heat map of correlation coefficients of Camel dataset

A7. CoreFX

The heat map of the correlation coefficients among the variables of interest in the CoreFX dataset is shown in Fig. A7.

Table A7 shows the statistics that can help to understand the distribution and variability of the different variables in the CoreFX dataset.

Table A6
Descriptive statistics of the model variables for Camel: defect-inducing commits after removal of the right-censored tail ($n = 5,471$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	24.79	16.00	22.83	(24.18, 25.39)
`entropy`	Treatment	1.84	1.67	1.39	(1.80, 1.88)
`sexp`	Treatment	4,645.5	1,709.0	6,523.7	(4,472.6, 4,818.3)
`rexp`	Treatment	31.18	1.02	415.33	(20.18, 42.19)
`days_to_first_fix`	Outcome	262.89	42.71	472.05	(250.38, 275.40)
`exp`	Confounder	12,543	6,444.0	15,277	(12,138, 12,948)
`ns`	Confounder	1.35	1.00	0.736	(1.33, 1.37)
`nf`	Confounder	10.12	5.00	32.00	(9.27, 10.97)
`lt`	Confounder	184.72	126.33	221.41	(178.85, 190.59)
`nuc`	Confounder	139.33	37.00	363.67	(129.69, 148.97)
`la`	Confounder	273.19	103.00	703.16	(254.56, 291.83)
`fix`	Confounder	0.177	0.00	0.382	(0.167, 0.187)
`age`	Mediator	65.51	17.92	129.92	(62.07, 68.95)

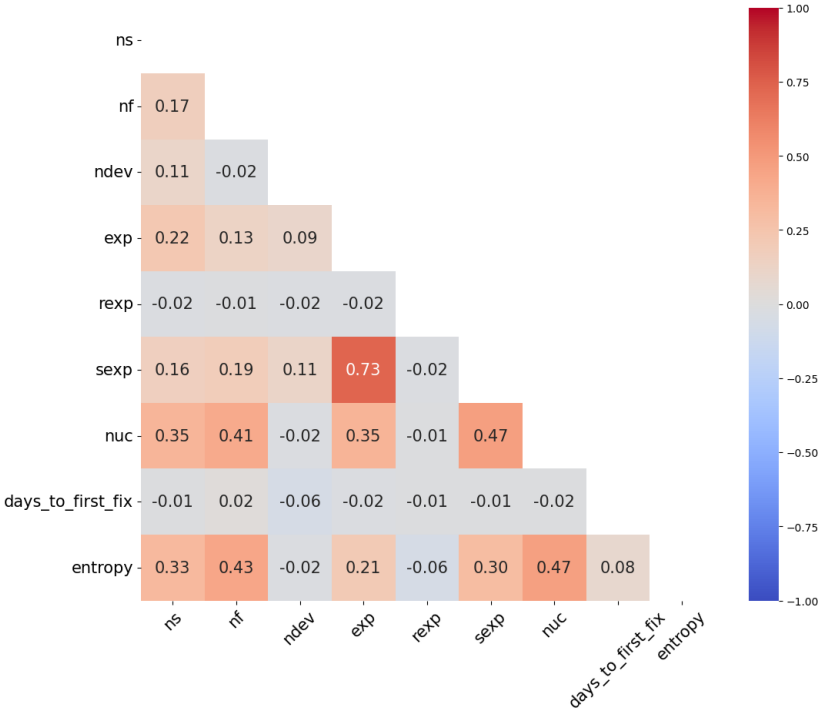


Figure A7: Heat map of correlation coefficients of CoreFX dataset

A8. Django

The heat map of the correlation coefficients among the variables of interest in the Django dataset is shown in Fig. A8.

Table A8 shows the statistics that can help to understand the distribution and variability of the different variables in the Django dataset.

Table A7
Descriptive statistics of the model variables for CoreFX: defect-inducing commits after removal of the right-censored tail ($n = 1,286$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	131.78	130.00	86.34	(127.06, 136.50)
`entropy`	Treatment	2.36	1.95	2.09	(2.25, 2.48)
`sexp`	Treatment	2,660.0	718.00	5,945.3	(2,335.0, 2,984.9)
`rexp`	Treatment	5.07	1.03	65.19	(1.50, 8.63)
`days_to_first_fix`	Outcome	233.97	93.14	301.93	(217.47, 250.48)
`exp`	Confounder	2,805.7	846.75	5,986.9	(2,478.5, 3,132.9)
`ns`	Confounder	1.19	1.00	0.491	(1.16, 1.21)
`nf`	Confounder	40.85	6.50	224.77	(28.56, 53.13)
`lt`	Confounder	458.83	223.75	2,057.0	(346.40, 571.26)
`nuc`	Confounder	599.73	58.00	3,518.1	(407.45, 792.01)
`la`	Confounder	3,009.1	108.00	16,285	(2,119.0, 3,899.2)
`fix`	Confounder	0.337	0.00	0.473	(0.311, 0.363)
`age`	Mediator	31.11	11.70	50.52	(28.35, 33.87)

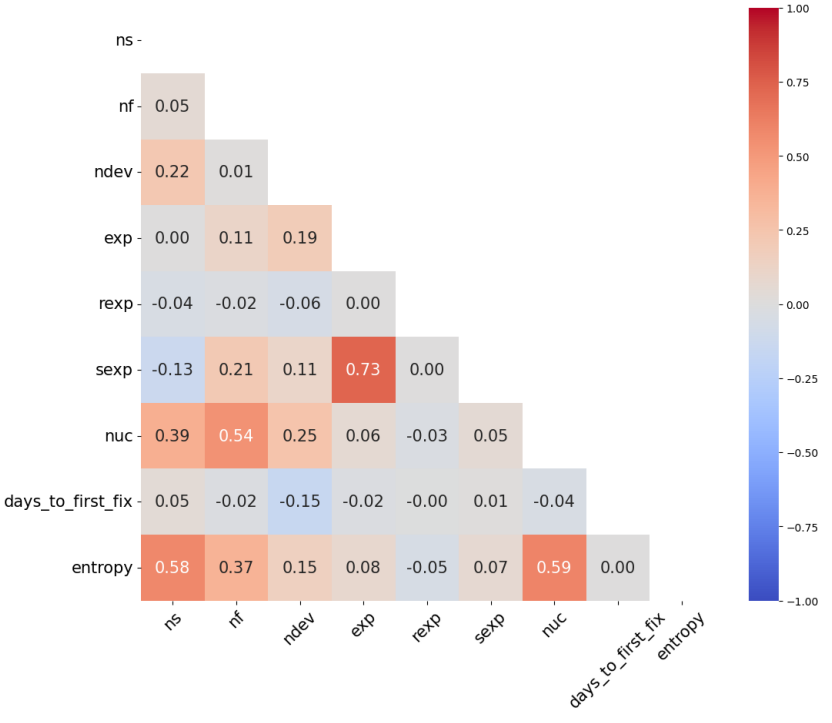


Figure A8: Heat map of correlation coefficients of Django dataset

A9. Nova

The heat map of the correlation coefficients among the variables of interest in the Nova dataset is shown in Fig. A9.

Table A9 shows the statistics that can help to understand the distribution and variability of the different variables in the Nova dataset.

Table A8
Descriptive statistics of the model variables for Django: defect-inducing commits after removal of the right-censored tail ($n = 10,510$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	48.19	26.00	51.18	(47.21, 49.17)
`entropy`	Treatment	1.04	0.943	1.04	(1.02, 1.06)
`sexp`	Treatment	717.30	312.00	1,152.8	(695.26, 739.34)
`rexp`	Treatment	34.57	1.04	390.42	(27.10, 42.03)
`days_to_first_fix`	Outcome	371.30	165.97	522.27	(361.31, 381.28)
`exp`	Confounder	2,095.8	892.75	2,714.1	(2,043.9, 2,147.7)
`ns`	Confounder	1.73	2.00	0.792	(1.72, 1.75)
`nf`	Confounder	4.63	2.00	12.96	(4.38, 4.88)
`lt`	Confounder	514.78	377.67	493.81	(505.34, 524.22)
`nuc`	Confounder	168.63	63.00	335.33	(162.22, 175.04)
`la`	Confounder	100.37	20.00	672.29	(87.52, 113.22)
`fix`	Confounder	0.731	1.00	0.444	(0.722, 0.739)
`age`	Mediator	48.91	18.66	102.46	(46.95, 50.86)

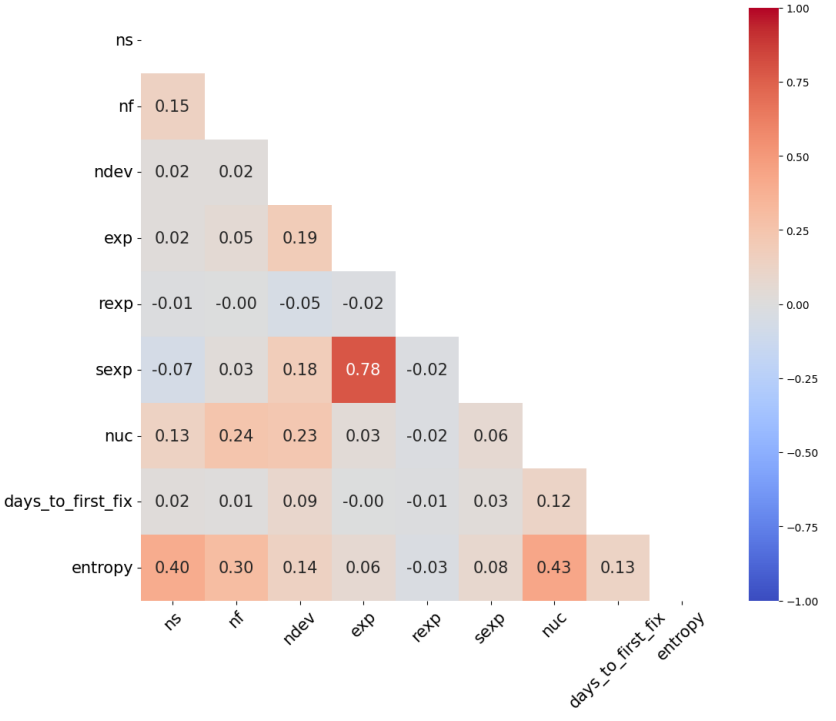


Figure A9: Heat map of correlation coefficients of Nova dataset

A10. JGroup

The heat map of the correlation coefficients among the variables of interest in the JGroup data set is shown in Fig. A10.

Table A10 shows the statistics that can help to understand the distribution and variability of the different variables in the JGroup dataset.

Table A9
Descriptive statistics of the model variables for Nova: defect-inducing commits after removal of the right-censored tail ($n = 11,011$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	158.45	144.00	104.23	(156.50, 160.39)
`entropy`	Treatment	1.18	0.984	1.10	(1.16, 1.20)
`sexp`	Treatment	389.54	126.00	699.96	(376.46, 402.61)
`rexp`	Treatment	125.86	1.05	2,762.6	(74.26, 177.46)
`days_to_first_fix`	Outcome	212.28	97.99	298.64	(206.71, 217.86)
`exp`	Confounder	514.00	175.00	892.23	(497.34, 530.67)
`ns`	Confounder	1.19	1.00	0.511	(1.18, 1.20)
`nf`	Confounder	5.94	2.00	46.50	(5.07, 6.81)
`lt`	Confounder	1,278.9	643.88	1,749.1	(1,246.3, 1,311.6)
`nuc`	Confounder	493.07	181.00	893.13	(476.39, 509.75)
`la`	Confounder	146.56	35.00	2,464.0	(100.54, 192.58)
`fix`	Confounder	0.368	0.00	0.482	(0.359, 0.377)
`age`	Mediator	13.52	3.65	47.04	(12.64, 14.40)

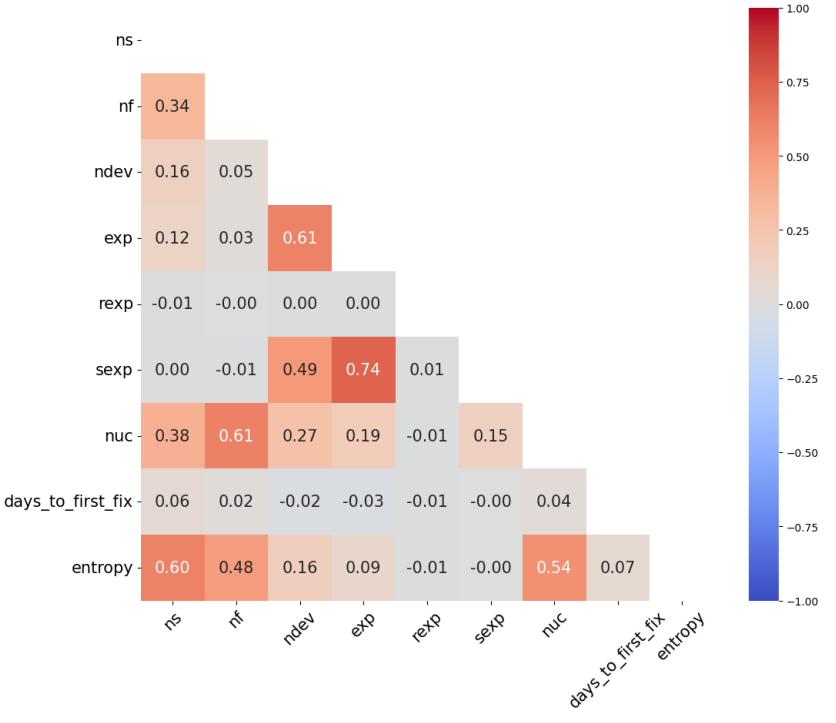


Figure A10: Heat map of correlation coefficients of JGroup dataset

A11. Fabric 8

The heat map of the correlation coefficient among the variables of our interest for the fabric data set is shown in Fig. A11.

Table A11 shows the statistics that can help to understand the distribution and variability of the different variables in the fabric dataset.

Table A10
Descriptive statistics of the model variables for JGroup: defect-inducing commits after removal of the right-censored tail ($n = 3,100$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes. The causal graph is rejected for this project (Section 4.1), so it is excluded from estimation; the statistics are reported for completeness.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	10.07	11.00	5.84	(9.87, 10.28)
`entropy`	Treatment	0.768	0.00	1.18	(0.726, 0.809)
`sexp`	Treatment	3,134.7	2,484.0	2,743.2	(3,038.1, 3,231.2)
`rexp`	Treatment	8.94	0.00	85.42	(5.93, 11.94)
`days_to_first_fix`	Outcome	421.46	114.72	648.50	(398.64, 444.29)
`exp`	Confounder	6,964.1	6,130.0	5,555.5	(6,768.5, 7,159.7)
`ns`	Confounder	1.31	1.00	0.649	(1.28, 1.33)
`nf`	Confounder	4.83	1.00	14.97	(4.30, 5.36)
`lt`	Confounder	1,409.3	1,034.9	1,438.6	(1,358.6, 1,459.9)
`nuc`	Confounder	289.66	81.00	708.97	(264.71, 314.62)
`la`	Confounder	132.53	39.00	313.06	(121.51, 143.55)
`fix`	Confounder	0.102	0.00	0.302	(0.091, 0.112)
`age`	Mediator	25.05	0.00	69.79	(22.59, 27.51)

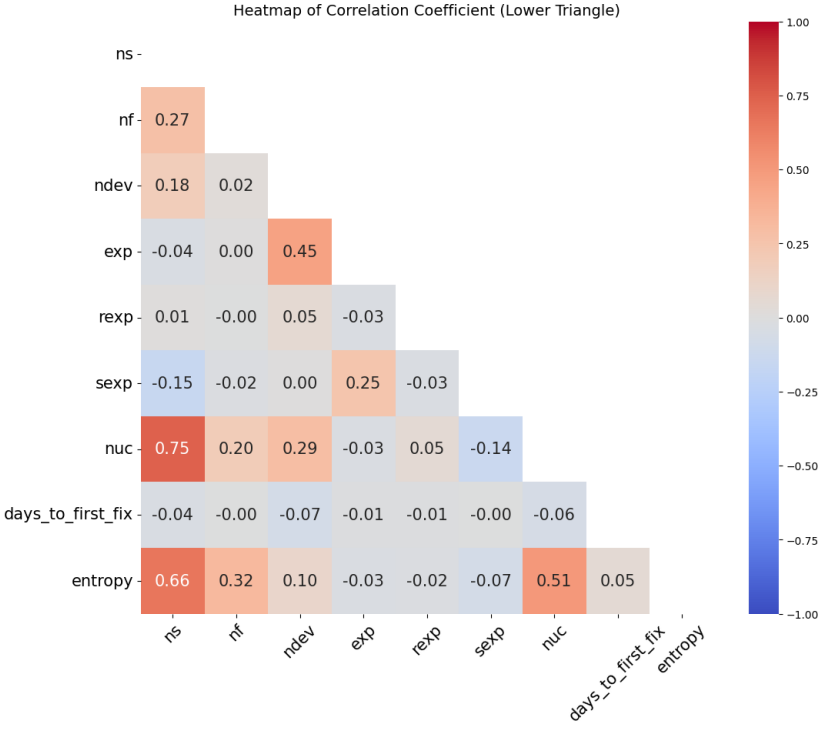


Figure A11: Heat map of correlation coefficients of Fabric 8 dataset

A12. Broadleaf

The heat map of the correlation coefficient among the variables of our interest for the Broadleaf data set is shown in Fig. A12.

Table A12 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table A11
Descriptive statistics of the model variables for Fabric8: defect-inducing commits after removal of the right-censored tail ($n = 2,561$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	32.95	31.00	16.15	(32.32, 33.57)
`entropy`	Treatment	1.55	1.33	1.48	(1.49, 1.61)
`sexp`	Treatment	1,139.1	417.00	1,554.6	(1,078.9, 1,199.3)
`rexp`	Treatment	15.00	1.15	166.14	(8.56, 21.43)
`days_to_first_fix`	Outcome	124.27	40.01	185.62	(117.08, 131.46)
`exp`	Confounder	8,777.9	5,745.5	9,014.3	(8,428.7, 9,127.0)
`ns`	Confounder	1.36	1.00	1.13	(1.32, 1.41)
`nf`	Confounder	12.91	3.00	72.74	(10.09, 15.72)
`lt`	Confounder	263.77	181.00	288.31	(252.60, 274.94)
`nuc`	Confounder	146.34	26.00	517.36	(126.30, 166.38)
`la`	Confounder	439.33	62.00	3,540.1	(302.22, 576.43)
`fix`	Confounder	0.204	0.00	0.403	(0.188, 0.219)
`age`	Mediator	15.66	3.36	34.51	(14.33, 17.00)

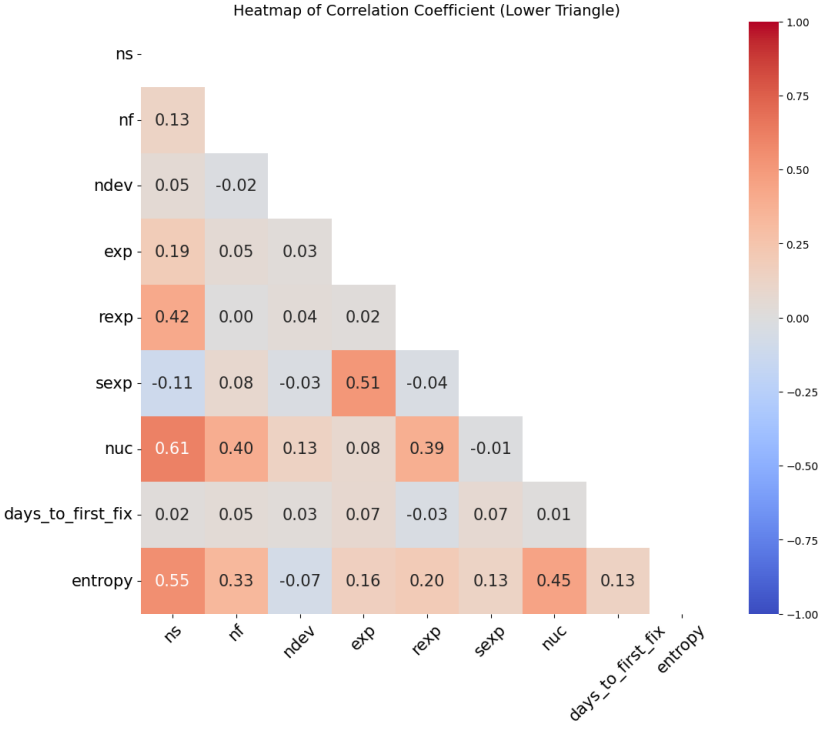


Figure A12: Heat map of correlation coefficients of Broadleaf dataset

A13. Brackets

The heat map of the correlation coefficient among the variables of our interest for the Brackets data set is shown in Fig. A13.

Table A13 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table A12
Descriptive statistics of the model variables for Broadleaf: defect-inducing commits after removal of the right-censored tail ($n = 1,990$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes. The causal graph is rejected for this project (Section 4.1), so it is excluded from estimation; the statistics are reported for completeness.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	20.89	24.00	6.39	(20.61, 21.17)
`entropy`	Treatment	1.75	1.53	1.61	(1.68, 1.82)
`sexp`	Treatment	2,486.6	742.00	5,298.6	(2,253.8, 2,719.4)
`rexp`	Treatment	12.76	1.07	109.78	(7.94, 17.58)
`days_to_first_fix`	Outcome	279.72	50.01	427.71	(260.92, 298.51)
`exp`	Confounder	5,795.7	2,526.8	7,724.1	(5,456.3, 6,135.1)
`ns`	Confounder	1.33	1.00	0.711	(1.30, 1.36)
`nf`	Confounder	22.33	4.00	118.34	(17.13, 27.53)
`lt`	Confounder	268.92	193.41	323.93	(254.69, 283.15)
`nuc`	Confounder	143.79	39.00	695.75	(113.22, 174.36)
`la`	Confounder	643.52	82.00	3,676.5	(481.98, 805.05)
`fix`	Confounder	0.230	0.00	0.421	(0.211, 0.248)
`age`	Mediator	25.51	7.64	47.12	(23.44, 27.58)

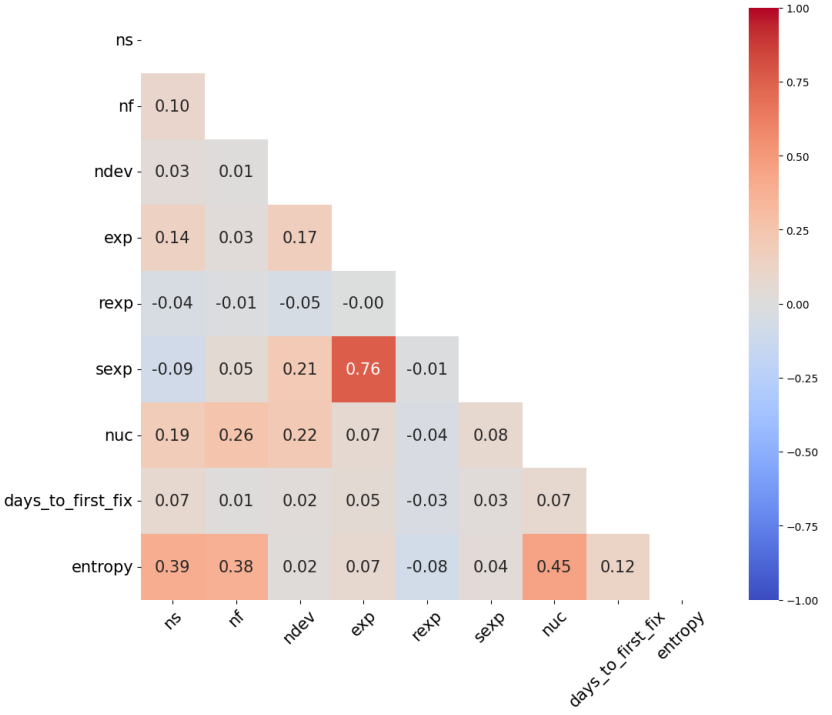


Figure A13: Heat map of correlation coefficients of Brackets dataset

B. Distributions of Selected Variables

Figures B1– B4 show the distributions of four variables in each of the thirteen projects: the number of past contributors to the touched files (`ndev`), the number of files changed (`nf`), the author's overall experience (`exp`) and defect lifetime (`days_to_first_fix`). Like Tables A1– A13, they describe the sample used for estimation, namely defect-inducing commits after removal of each project's right-censored tail (Section 2.2), not all commits in the source

Table A13
Descriptive statistics of the model variables for Brackets: defect-inducing commits after removal of the right-censored tail ($n = 3,860$). CI: 95% confidence interval for the mean. `fix` is binary, so its mean is the share of defect-inducing commits that are themselves fixes.

Variable	Role	Mean	Median	SD	95% CI
`ndev`	Treatment	37.84	34.00	23.95	(37.09, 38.60)
`entropy`	Treatment	0.759	0.578	0.885	(0.731, 0.787)
`sexp`	Treatment	371.26	173.00	496.88	(355.58, 386.93)
`rexp`	Treatment	19.27	1.60	87.38	(16.51, 22.03)
`days_to_first_fix`	Outcome	174.70	14.95	343.56	(163.86, 185.54)
`exp`	Confounder	751.92	333.75	1,060.6	(718.46, 785.37)
`ns`	Confounder	1.21	1.00	0.422	(1.19, 1.22)
`nf`	Confounder	3.52	2.00	11.28	(3.16, 3.87)
`lt`	Confounder	640.44	487.36	1,282.8	(599.98, 680.91)
`nuc`	Confounder	178.86	73.00	391.36	(166.51, 191.20)
`la`	Confounder	175.69	31.50	1,864.9	(116.85, 234.52)
`fix`	Confounder	0.262	0.00	0.440	(0.248, 0.276)
`age`	Mediator	10.93	1.37	33.41	(9.87, 11.98)

corpus. Commits that introduced no defect have no defect lifetime, and including them, would place a large mass of observations at zero days that does not correspond to any defect.

Each figure is a violin plot: a kernel density estimate of the variable, mirrored about a vertical axis, so that wider sections mark values where commits are more concentrated. Dashed lines mark the quartiles. The density is drawn only over the observed range of each variable, so no part of a violin extends below the smallest value recorded. Because `nf`, `exp` and defect lifetime span several orders of magnitude, Figs. B.2–B.4 plot $\log_{10}(1+x)$ and label the axis in the original units; `ndev`, whose range is narrower, is shown on a linear scale. Violins are scaled to the same maximum width, so their widths compare shapes, not sample sizes; the number of commits in each project is given in Tables A1– A13. JGroup and Broadleaf, for which the causal graph is rejected (Section 4.1), are shown in grey for completeness.

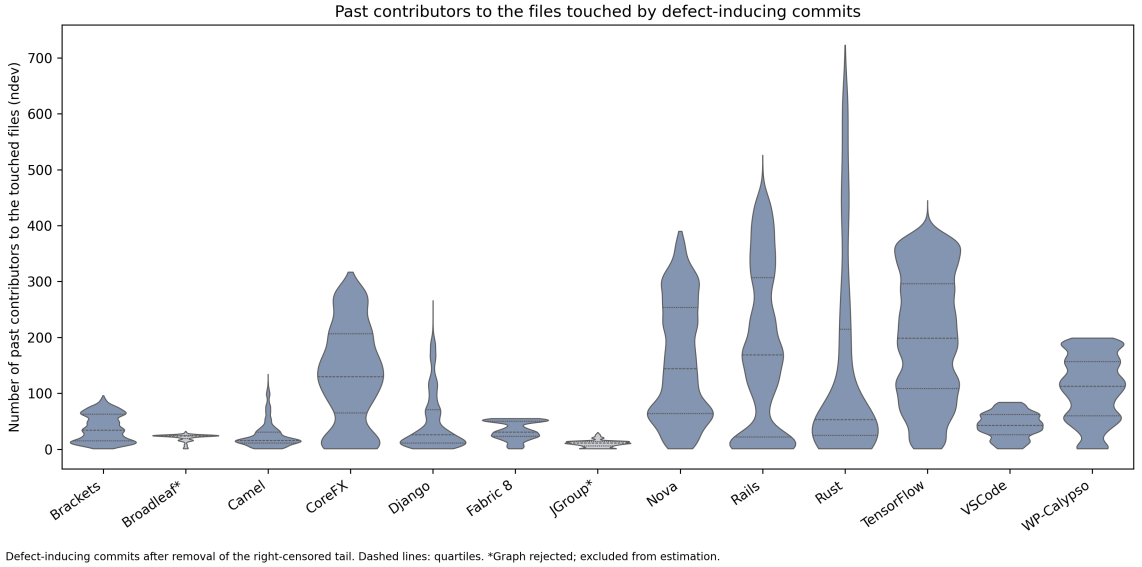


Figure B1: Number of past contributors (ndev)

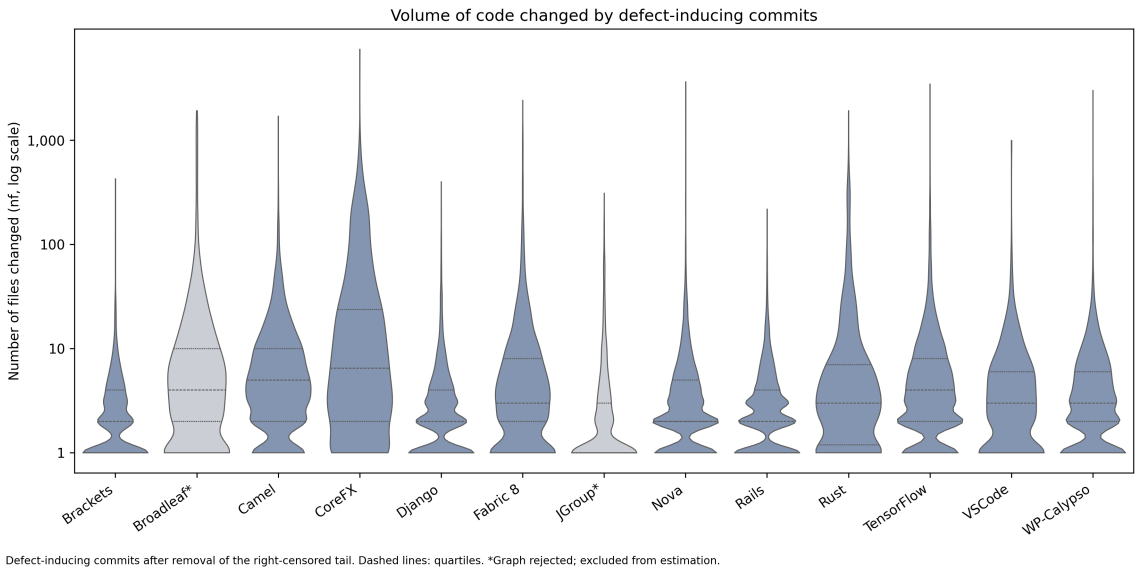


Figure B2: Number of files changed (nf)

Number of past contributors (ndev)

The number of developers who had previously modified the files a commit touches varies more between projects than within most of them. Medians range from 11 in JGroup and 16 in Camel to 169 in Rails and 199 in TensorFlow. In most projects the distribution is roughly symmetric, but in Rust, Django and Camel it is long-tailed: Rust's median is 53, yet a minority of its commits touch files with more than 600 past contributors. JGroup, Broadleaf and Fabric8 are tightly concentrated, with standard deviations below 17. This variation in scale is one reason we interpret the effect of ndev per standard deviation as well as per developer (Table 5).

Number of files changed (nf)

Most defect-inducing commits are small. The median commit touches between one file (JGroup) and six or seven (CoreFX), and in every project the interquartile range lies below about 25 files. Every distribution nonetheless has a long upper tail of commits touching hundreds or thousands of files, most visibly in CoreFX. Because nf confounds the effect of entropy and mediates the effect of recent experience (Section 2.4), this tail matters for estimation: entropy is closely tied to the number of files changed, and Section 4.2 shows that several entropy estimates depend on the most widely dispersed commits.

Overall experience (exp)

Author experience, measured as the number of prior commits, differs by two orders of magnitude between projects. Median experience is 175 commits in Nova and around 300 in Brackets and WP-Calypso, against 5,700 to 6,400 in Fabric8 and Camel. Within projects, the distribution is strongly right-skewed. TensorFlow is bimodal: most authors of defect-inducing commits have fewer than a few thousand prior commits, but a distinct group has close to 100,000. This group, which likely reflects automated or highly concentrated commit activity, produces TensorFlow's extreme mean-to-median ratio for subsystem experience (Table C3). Because exp confounds the effect of recent experience, it enters that adjustment set directly (Section 2.4).

C. Summary statistics

C1. Summary statistics of developer count across all 13 project datasets
C2. Summary statistics of entropy across all 13 project datasets
C3. Summary statistics of *subsystem experience* across all 13 project datasets.
C4. Summary statistics of *recent experience* across all 13 project datasets.

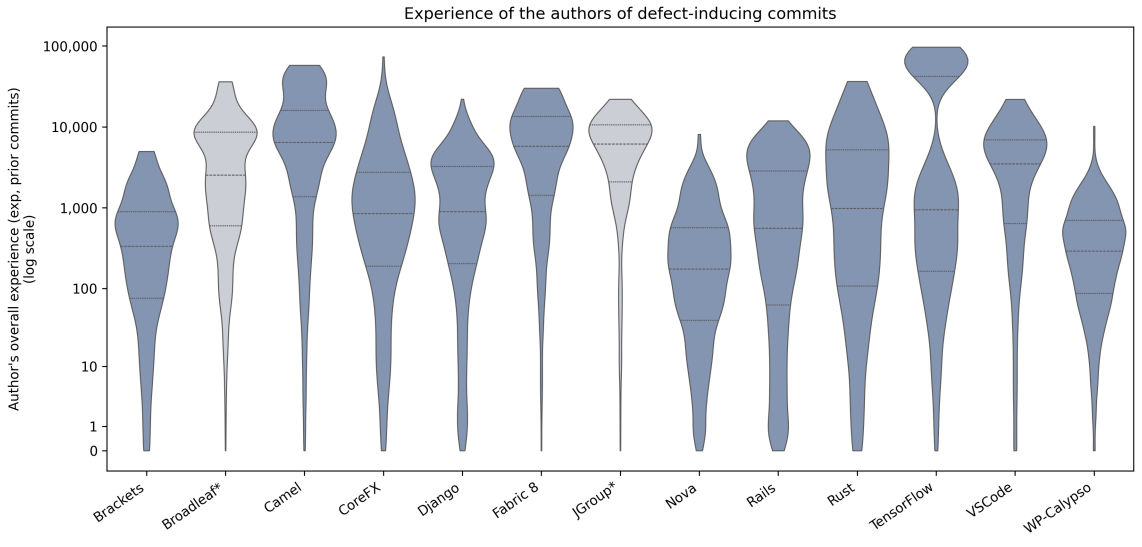


Figure B3: Overall experience (exp)

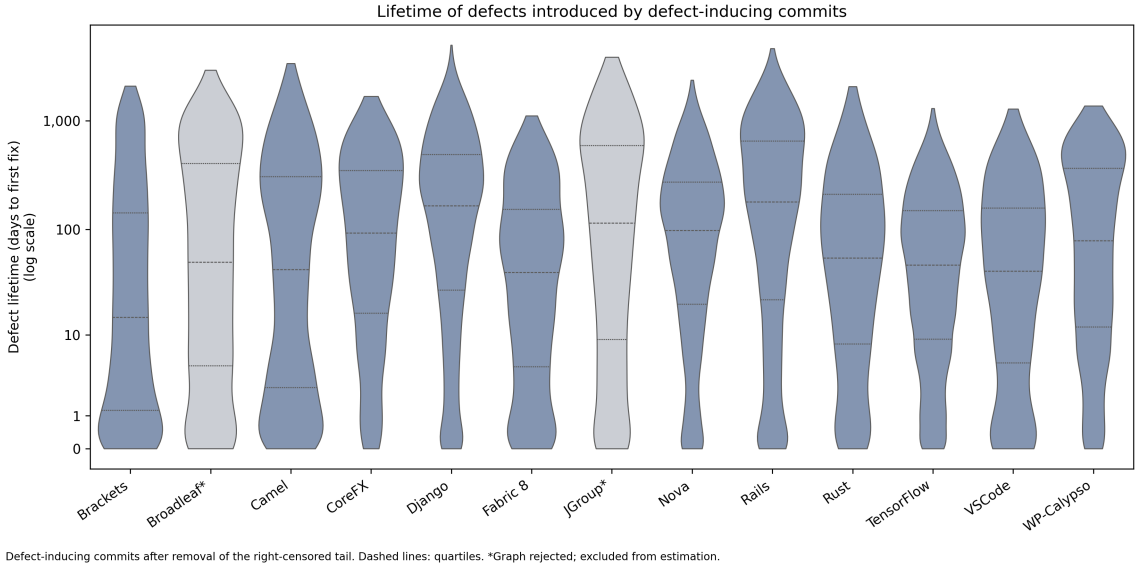


Figure B4: Defect lifetime

D. Falsification of the causal graph

A causal graph is a set of claims about the data, and some of them can be checked. Every absent edge implies a conditional independence: under the local Markov condition, each variable should be independent of its non-descendants once its parents are known. If the data contradict many of these implications, the graph cannot be the process that produced them, and estimates built on it are unreliable. This section reports how the graph of Fig. 2 fares on each of the 13 projects. Passing the test does not show that the graph is correct; it shows only that the data fail to contradict it.

Procedure. We used the permutation-based procedure of Eulig et al.[1], implemented in DoWhy's gcm.falsify module, in four steps.

Project	n	Mean	Median	SD	Mean/median	Skew
Brackets	3,860	37.84	34.00	23.95	1.11	Right (0.481)
Broadleaf*	1,990	20.89	24.00	6.39	0.87	Left (−1.459)
Camel	5,471	24.79	16.00	22.83	1.55	Right (1.155)
CoreFX	1,286	131.78	130.00	86.34	1.01	Right (0.062)
Django	10,510	48.19	26.00	51.18	1.85	Right (1.301)
Fabric8	2,561	32.95	31.00	16.15	1.06	Right (0.362)
JGroup*	3,100	10.07	11.00	5.84	0.92	Left (−0.478)
Nova	11,011	158.45	144.00	104.23	1.10	Right (0.416)
Rails	13,086	182.56	169.00	140.41	1.08	Right (0.290)
Rust	1,422	145.89	53.00	178.56	2.75	Right (1.561)
TensorFlow	12,926	201.63	199.00	111.43	1.01	Right (0.071)
VSCode	1,023	42.85	43.00	22.67	1.00	Left (−0.019)
WP-Calypso	5,596	108.30	113.00	57.41	0.96	Left (−0.246)

Table C1
Summary statistics of number of past contributors to the touched files (`ndev`) across the 13 projects, computed on defect-inducing commits after removal of the right-censored tail. Skew is Pearson's second coefficient, $3(\text{mean} - \text{median})/\text{SD}$. Mean/median is shown because for strongly skewed variables it is a better guide to shape than the normalised coefficient. *Graph rejected; excluded from estimation.

Project	n	Mean	Median	SD	Mean/median	Skew
Brackets	3,860	0.759	0.578	0.885	1.31	Right (0.613)
Broadleaf*	1,990	1.75	1.53	1.61	1.14	Right (0.409)
Camel	5,471	1.84	1.67	1.39	1.10	Right (0.362)
CoreFX	1,286	2.36	1.95	2.09	1.21	Right (0.590)
Django	10,510	1.04	0.943	1.04	1.10	Right (0.271)
Fabric8	2,561	1.55	1.33	1.48	1.16	Right (0.439)
JGroup*	3,100	0.768	0.00	1.18	—	Right (1.955)
Nova	11,011	1.18	0.984	1.10	1.20	Right (0.541)
Rails	13,086	1.07	0.977	0.976	1.09	Right (0.280)
Rust	1,422	1.47	1.09	1.55	1.35	Right (0.727)
TensorFlow	12,926	1.51	1.31	1.25	1.16	Right (0.496)
VSCode	1,023	1.23	1.03	1.15	1.19	Right (0.518)
WP-Calypso	5,596	1.32	1.15	1.15	1.15	Right (0.444)

Table C2
Summary statistics of change entropy (`entropy`) across the 13 projects, computed on defect-inducing commits after removal of the right-censored tail. Skew is Pearson's second coefficient, $3(\text{mean} - \text{median})/\text{SD}$. Mean/median is shown because for strongly skewed variables it is a better guide to shape than the normalised coefficient. *Graph rejected; excluded from estimation.

1. **Test the graph's implications**. For each node, we tested its independence from every non-descendant given its parents, using a significance level of 0.05. The graph of Fig. 2 implies 66 such statements. We used a partial-correlation test on ranks (a Spearman partial correlation). It is robust to the heavy tails of la, lt and nuc and remained practical at our sample sizes, unlike kernel-based tests.
2. **Count violations**. We recorded the share of the 66 statements the data reject.
3. **Build a baseline**. A raw count of violations is hard to read on its own. With several thousand commits per project, a rank test detects even very weak dependences, so most graphs over these variables violate a large share of their implications. The procedure therefore compares the proposed graph with 1,000 graphs that keep its structure but randomly permute which variable sits at which node. Each permuted graph is tested in the same way.
4. **Compute two quantities.**
   - **p_LMC** is the share of permuted graphs that violate no greater share of their implications than the proposed graph does. A small value means the proposed graph fits the data better than almost all structurally identical

Project	n	Mean	Median	SD	Mean/median	Skew
Brackets	3,860	371.26	173.00	496.88	2.15	Right (1.197)
Broadleaf*	1,990	2,486.6	742.00	5,298.6	3.35	Right (0.988)
Camel	5,471	4,645.5	1,709.0	6,523.7	2.72	Right (1.350)
CoreFX	1,286	2,660.0	718.00	5,945.3	3.70	Right (0.980)
Django	10,510	717.30	312.00	1,152.8	2.30	Right (1.055)
Fabric8	2,561	1,139.1	417.00	1,554.6	2.73	Right (1.393)
JGroup*	3,100	3,134.7	2,484.0	2,743.2	1.26	Right (0.712)
Nova	11,011	389.54	126.00	699.96	3.09	Right (1.130)
Rails	13,086	525.91	129.00	798.93	4.08	Right (1.490)
Rust	1,422	3,970.4	910.00	6,796.9	4.36	Right (1.351)
TensorFlow	12,926	19,009	817.00	30,373	23.27	Right (1.797)
VSCode	1,023	2,415.1	552.00	3,872.3	4.38	Right (1.443)
WP-Calypso	5,596	421.45	201.00	651.75	2.10	Right (1.015)

Table C3
Summary statistics of subsystem experience (`sexp`) across the 13 projects, computed on defect-inducing commits after removal of the right-censored tail. Skew is Pearson's second coefficient, $3(\text{mean} - \text{median})/\text{SD}$. Mean/median is shown because for strongly skewed variables it is a better guide to shape than the normalised coefficient. *Graph rejected; excluded from estimation.

Project	n	Mean	Median	SD	Mean/median	Skew
Brackets	3,860	19.27	1.60	87.38	12.04	Right (0.607)
Broadleaf*	1,990	12.76	1.07	109.78	11.97	Right (0.320)
Camel	5,471	31.18	1.02	415.33	30.48	Right (0.218)
CoreFX	1,286	5.07	1.03	65.19	4.94	Right (0.186)
Django	10,510	34.57	1.04	390.42	33.33	Right (0.258)
Fabric8	2,561	15.00	1.15	166.14	13.09	Right (0.250)
JGroup*	3,100	8.94	0.00	85.42	—	Right (0.314)
Nova	11,011	125.86	1.05	2,762.6	119.66	Right (0.136)
Rails	13,086	38.53	1.12	442.19	34.38	Right (0.254)
Rust	1,422	41.16	1.10	332.96	37.57	Right (0.361)
TensorFlow	12,926	7.94	1.09	195.88	7.25	Right (0.105)
VSCode	1,023	24.50	1.23	150.33	19.88	Right (0.464)
WP-Calypso	5,596	27.59	1.07	241.86	25.75	Right (0.329)

Table C4
Summary statistics of recent experience (`rexp`) across the 13 projects, computed on defect-inducing commits after removal of the right-censored tail. Skew is Pearson's second coefficient, $3(\text{mean} - \text{median})/\text{SD}$. Mean/median is shown because for strongly skewed variables it is a better guide to shape than the normalised coefficient. *Graph rejected; excluded from estimation.

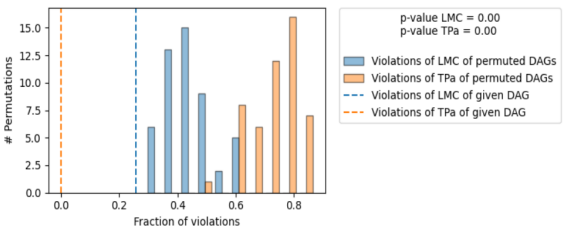
alternatives. A large value means it does no better than a random relabelling. Following DoWhy's default, we reject the graph for a project when $p_LMC > 0.05$.

- **p_TPa** is the share of permuted graphs that fall in the proposed graph's Markov equivalence class. When it is small, the test is informative: the data could, in principle, distinguish the proposed graph from its permutations. A test that cannot do so would be uninformative, whatever its p_LMC.

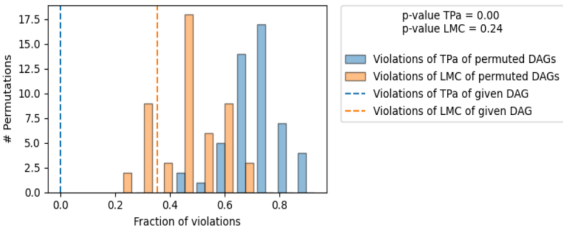
To check that verdicts did not depend on the random permutations drawn, we repeated the test with three random seeds. Because every test is a partial correlation, all of them can be computed from one rank-correlation matrix per project. We used this to run the 39,000 permutations efficiently, and confirmed the verdicts with DoWhy's own implementation.

Results: Figures D1 and D2 show histograms of LMC violations for permuted DAGs and TPa violations, and each panel's legend shows the colour scheme.

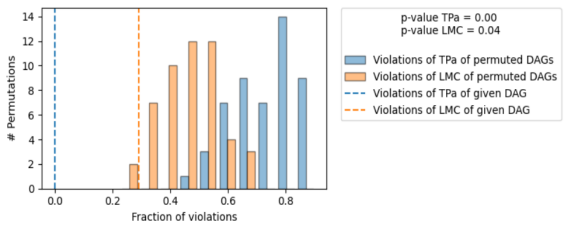
- **Informative everywhere.** No permuted graph fell in the proposed graph's equivalence class in any project ($p_{\text{TPa}} = 0.000$).
- **Not rejected in 11 projects.** Brackets, Camel, CoreFX, Django, Fabric8, Nova, Rails, Rust, TensorFlow, VSCode and WP-Calypso all have p_LMC between 0.001 and 0.04 across seeds. In each of them, the proposed graph violates fewer of its implications than at least 96% of its permutations.
- **Rejected in two projects.** For Broadleaf (p_LMC 0.24–0.29) and JGroup (0.22–0.24), the graph fits no better than a random relabelling of its variables.
- **Robust verdicts.** The two groups are far apart. Any cutoff between 0.04 and 0.31 yields the same verdicts, and all three seeds agree on every project.



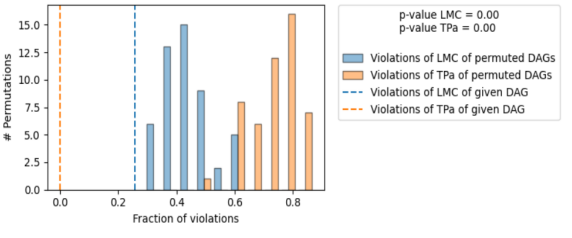
(a) Brackets



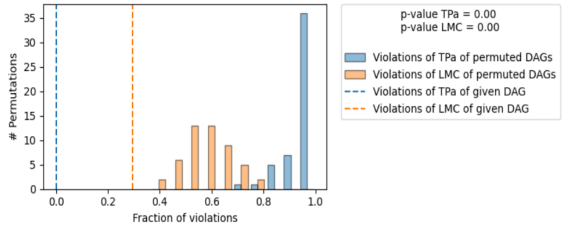
(b) Broadleaf



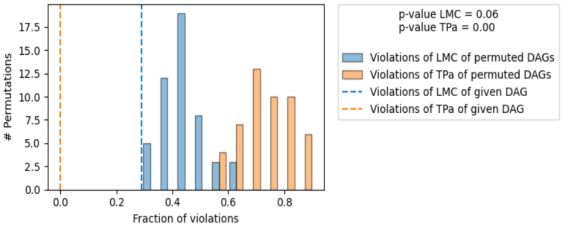
(c) Camel



(d) CoreFX



(e) Django



(f) Fabric 8

Figure D1: DAG falsification summary of (a) Brackets, (b) Broadleaf, (c) Camel, (d) CoreFX, (e) Django, and (f) Fabric 8.

E. Causal-effect estimates

E1. Causal effect of ndev on defect lifetimee

Table E1 summarises the causal inference results for developer count on defect lifetime across different software.

E2. Causal effect of entropy on defect lifetime

Table E2 summarises the causal inference results for the effect of *entropy* on defect lifetime across different software.

E3. Causal effect of sexp on defect lifetime

Table E3 summarises the causal inference results for subsystem experience on defect lifetime across software.

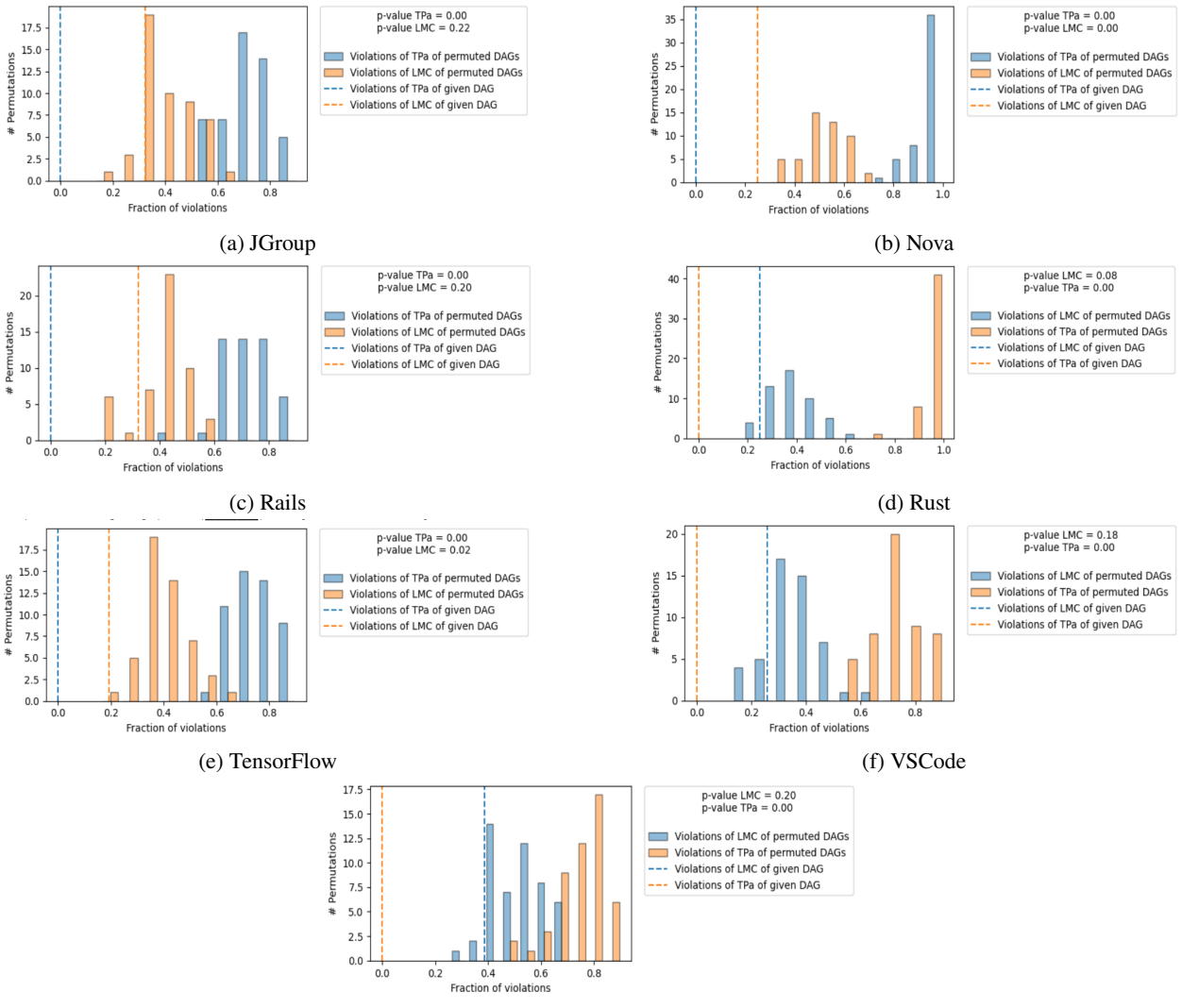


Figure D2: DAG falsification summary of (a) JGroup, (b) Nova, (c) Rails, (d) Rust, (e) TensorFlow, (f) VSCode and (g) WP-Calypso.

E4. Causal effect of rexp on defect lifetime

Table E4 summarises the causal inference results for rexp on days_to_first_fix.

References

[1] Eulig, E., Mastakouri, A.A., Blöbaum, P., Hardt, M., Janzing, D.: Toward falsifying causal graphs using a permutation-based test. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, no. 25, pp. 26778–26786 (2025)

Table E1
Causal effect of `ndev` on defect lifetime: days per additional developer (`ndev`) across the 11 projects compatible with the causal graph. Estimates are back-door-adjusted linear regressions on the adjustment set {time, `lt`, `nuc`}, fitted to defect-inducing commits after removal of the right-censored tail; n is the number of commits analysed. 95% CIs and p-values use HC3 heteroskedasticity-robust standard errors; bold p-values are below 0.05. RCC: estimate after adding a simulated random common cause; DSR: mean estimate over random 80% subsamples; PT: estimate after replacing the treatment with a permuted placebo (50 simulations each; no refuter rejected any estimate, all refuter $p \geq 0.31$).

		Estimate			Refutation		
Project	n	Effect	95% CI	p	RCC	DSR	PT
Brackets	3,860	−0.0682	[−1.501, 1.365]	0.926	−0.0676	−0.0275	−0.00121
Camel	5,471	−2.408	[−3.064, −1.751]	**<0.001**	−2.406	−2.420	0.0221
CoreFX	1,286	−0.312	[−0.538, −0.0863]	**0.007**	−0.312	−0.323	−0.0244
Django	10,510	0.174	[−0.160, 0.508]	0.308	0.174	0.189	0.000700
Fabric8	2,561	1.556	[0.556, 2.555]	**0.002**	1.555	1.565	−0.0523
Nova	11,011	0.596	[0.381, 0.810]	**<0.001**	0.596	0.593	0.000937
Rails	13,086	−0.900	[−1.169, −0.630]	**<0.001**	−0.900	−0.903	0.00836
Rust	1,422	−0.427	[−0.734, −0.120]	**0.006**	−0.427	−0.416	−0.00673
TensorFlow	12,926	−0.249	[−0.345, −0.153]	**<0.001**	−0.249	−0.255	−0.00244
VSCode	1,023	−1.097	[−3.568, 1.373]	0.384	−1.094	−1.035	0.0578
WP-Calypso	5,596	−0.865	[−1.170, −0.561]	**<0.001**	−0.865	−0.855	0.0108

Table E2
Causal effect of `entropy` on defect lifetime: days per unit of entropy (`entropy`) across the 11 projects compatible with the causal graph. Estimates are back-door-adjusted linear regressions on the adjustment set {`nf`, `la`, `nuc`}, fitted to defect-inducing commits after removal of the right-censored tail; n is the number of commits analysed. 95% CIs and p-values use HC3 heteroskedasticity-robust standard errors; bold p-values are below 0.05. RCC: estimate after adding a simulated random common cause; DSR: mean estimate over random 80% subsamples; PT: estimate after replacing the treatment with a permuted placebo (50 simulations each; no refuter rejected any estimate, all refuter $p \geq 0.31$).

		Estimate			Refutation		
Project	n	Effect	95% CI	p	RCC	DSR	PT
Brackets	3,860	5.92	[−16.60, 28.44]	0.606	5.89	5.14	0.16
Camel	5,471	−4.26	[−15.89, 7.37]	0.473	−4.26	−4.07	0.82
CoreFX	1,286	−10.43	[−20.84, −0.0056]	**0.0499**	−10.41	−9.28	−0.11
Django	10,510	−137.55	[−149.34, −125.75]	**<0.001**	−137.54	−138.01	−1.11
Fabric8	2,561	−10.08	[−15.55, −4.61]	**<0.001**	−10.07	−10.22	−0.056
Nova	11,011	−3.58	[−9.69, 2.54]	0.251	−3.58	−3.78	−0.17
Rails	13,086	−38.06	[−54.18, −21.93]	**<0.001**	−38.05	−38.99	−1.23
Rust	1,422	−1.70	[−14.45, 11.05]	0.794	−1.68	−1.42	0.11
TensorFlow	12,926	−14.43	[−17.65, −11.22]	**<0.001**	−14.44	−14.45	−0.0055
VSCode	1,023	−8.85	[−20.59, 2.89]	0.139	−8.85	−8.46	0.79
WP-Calypso	5,596	−31.83	[−40.45, −23.22]	**<0.001**	−31.85	−31.10	−0.0012

Table E3
Causal effect of `sexp` on defect lifetime: days per additional prior change to the subsystem (`sexp`) across the 11 projects compatible with the causal graph. Estimates are back-door-adjusted linear regressions on the adjustment set {`author_time`, `ns`}, fitted to defect-inducing commits after removal of the right-censored tail; n is the number of commits analysed. 95% CIs and p-values use HC3 heteroskedasticity-robust standard errors; bold p-values are below 0.05. RCC: estimate after adding a simulated random common cause; DSR: mean estimate over random 80% subsamples; PT: estimate after replacing the treatment with a permuted placebo (50 simulations each; no refuter rejected any estimate, all refuter $p \geq 0.31$).

		Estimate			Refutation		
Project	n	Effect	95% CI	p	RCC	DSR	PT
Brackets	3,860	0.0116	[−0.0126, 0.0357]	0.348	0.0116	0.0118	−0.00260
Camel	5,471	0.00444	[0.00279, 0.00609]	**<0.001**	0.00444	0.00436	0.0000401
CoreFX	1,286	0.00159	[−0.00128, 0.00446]	0.278	0.00159	0.00155	0.0000792
Django	10,510	0.00957	[0.000403, 0.0187]	**0.041**	0.00957	0.0101	−0.000310
Fabric8	2,561	−0.0152	[−0.0190, −0.0113]	**<0.001**	−0.0152	−0.0152	0.0000289
Nova	11,011	−0.00555	[−0.0143, 0.00325]	0.216	−0.00554	−0.00527	−0.000230
Rails	13,086	0.0107	[−0.00330, 0.0247]	0.134	0.0107	0.0110	−0.00209
Rust	1,422	−0.000327	[−0.00269, 0.00204]	0.786	−0.000332	−0.000298	−0.000268
TensorFlow	12,926	0.000289	[0.000201, 0.000376]	**<0.001**	0.000289	0.000283	-1.222×10^{-6}
VSCode	1,023	−0.000933	[−0.00343, 0.00157]	0.465	−0.000936	−0.000844	0.000113
WP-Calypso	5,596	0.00397	[−0.00531, 0.0133]	0.402	0.00397	0.00380	0.000306

Table E4
Causal effect of `rexp` on defect lifetime: days per unit of recent experience (`rexp`) across the 11 projects compatible with the causal graph. Estimates are back-door-adjusted linear regressions on the adjustment set {`author_time`, `exp`}, fitted to defect-inducing commits after removal of the right-censored tail; n is the number of commits analysed. 95% CIs and p-values use HC3 heteroskedasticity-robust standard errors; bold p-values are below 0.05. RCC: estimate after adding a simulated random common cause; DSR: mean estimate over random 80% subsamples; PT: estimate after replacing the treatment with a permuted placebo (50 simulations each; no refuter rejected any estimate, all refuter $p \geq 0.31$).

		Estimate			Refutation		
Project	n	Effect	95% CI	p	RCC	DSR	PT
Brackets	3,860	−0.138	[−0.224, −0.0527]	**0.002**	−0.138	−0.139	0.00791
Camel	5,471	−0.0203	[−0.0310, −0.00968]	**<0.001**	−0.0204	−0.0200	−0.00241
CoreFX	1,286	0.299	[−0.292, 0.890]	0.321	0.299	0.253	−0.0215
Django	10,510	0.0116	[−0.0196, 0.0428]	0.468	0.0116	0.0119	−0.00146
Fabric8	2,561	0.0408	[−0.0754, 0.157]	0.492	0.0408	0.0360	−0.00343
Nova	11,011	0.0000126	[−0.00111, 0.00113]	0.982	0.0000130	9.741×10^{-6}	−0.000179
Rails	13,086	−0.00230	[−0.0199, 0.0153]	0.798	−0.00229	−0.00159	−0.00165
Rust	1,422	0.00285	[−0.0241, 0.0298]	0.836	0.00287	0.00389	0.000773
TensorFlow	12,926	−0.00587	[−0.0215, 0.00978]	0.462	−0.00588	−0.00668	−0.00100
VSCode	1,023	−0.0128	[−0.0805, 0.0549]	0.711	−0.0128	−0.00820	0.00583
WP-Calypso	5,596	0.00978	[−0.0257, 0.0453]	0.589	0.00976	0.00951	−0.00221